

Automated Biometric System for Individual Re-Identification of Mugger Crocodiles Using Deep Learning

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Abstract—Individual re-identification (Re-ID) of wildlife species is a fundamental challenge in ecological monitoring and conservation research. Traditional approaches rely on invasive physical tagging, which introduces stress and logistical difficulty for large reptiles such as crocodilians. In this work, we present an automated biometric system for the individual re-identification of mugger crocodiles (*Crocodylus palustris*) based on their distinctive dorsal scute patterns, leveraging aerial imagery acquired via Unmanned Aerial Vehicles (UAVs). We investigate two deep learning architectures: Inception-v3 and YOLOv5 for this task. The dataset comprises approximately 4,094 training images and 32 test images spanning six known individual identities (classes 107–112), supplemented by an additional 32-sample evaluation set with unknown ground-truth labels. Through systematic experimentation, we identify and address methodological pitfalls including whole-body classification instead of scute-region focus, incorrect bounding box annotation strategies, dataset mismatch during evaluation, and class prediction bias. Our analysis demonstrates that the bounding box-guided YOLOv5 pipeline, trained exclusively on annotated scute regions, produces more discriminative feature representations than non-annotated classification approaches. We discuss the influence of illumination, occlusion, and inter-individual scute similarity on model performance, and propose directions for future improvement. This study contributes toward a scalable, non-invasive monitoring system applicable to vulnerable crocodilian populations in the wild.

Index Terms—Individual Re-Identification, Mugger Crocodile, Dorsal Scute Patterns, Convolutional Neural Networks, YOLOv5, Inception-v3, UAV Imagery, Wildlife Biometrics, Deep Learning, Conservation Technology

I. INTRODUCTION

Individual identification (IID) of animals is a cornerstone of modern ecological research, enabling fine-grained studies of behavior, physiology, movement, and population dynamics [1]. For species classified as vulnerable or endangered, the ability to track specific individuals non-invasively over time is particularly critical for guiding conservation interventions. However, most classical IID approaches: physical tagging, radio collaring, notching require capture and restraint, which is both logistically demanding and potentially harmful, particularly for large reptiles such as crocodilians [1].

Computer vision and deep learning have emerged as powerful alternatives, enabling identification from photographs or video without physical contact. Convolutional Neural Networks (CNNs) have demonstrated strong performance across diverse species including tigers [4], chimpanzees, birds, and cetaceans. A recurring theme across this literature is the importance of focusing model attention on biologically meaningful,

discriminative features: stripe patterns, facial landmarks, or, in the case of crocodilians, the arrangement of dorsal scutes.

The mugger crocodile (*Crocodylus palustris*), categorized as Vulnerable by the IUCN, inhabits a wide range of freshwater habitats across South Asia. India holds one of the largest wild populations, concentrated significantly in the state of Gujarat. Conservation efforts for this species are hindered by the animals' tendency to inhabit areas outside protected reserves and by ongoing human–wildlife conflict. Developing a robust automated Re-ID system tailored to mugger crocodiles thus has immediate practical value.

In this report, we describe the design, development, and evaluation of a deep learning-based Re-ID system for mugger crocodiles using UAV-acquired aerial imagery. Our system focuses on dorsal scute patterns as the primary biometric feature and employs two complementary architectures: Inception-v3 for image-level classification [2] and YOLOv5 for detection-guided classification [3].

II. LITERATURE REVIEW

The application of deep learning to wildlife biometrics has accelerated significantly over the past decade. Early work demonstrated that CNNs could outperform traditional hand-crafted feature methods for species classification [3], and subsequent studies extended these techniques to individual-level identification.

For crocodilians specifically, Balaguera-Reina et al. explored semi-automated IID of American crocodiles using manually coded dorsal scute arrangements, establishing that scute placement constitutes a unique, stable biometric identifier across individuals. Desai et al. [1] built upon this insight with a CNN-based system trained on 88,000 UAV images from 143 free-ranging mugger crocodile individuals across 19 locations in Gujarat, India, demonstrating that YOLO-v5l with annotated bounding boxes achieved a True Positive Rate (TPR) of 88.8% and a True Negative Rate (TNR) of 89.6% at a decision threshold of 0.84.

The Inception-v3 architecture [2] introduced factorized convolutions and auxiliary classifiers to improve both computational efficiency and classification accuracy on large-scale datasets. However, its limitation in handling spatially-annotated inputs makes it less suitable for localized feature detection. In contrast, YOLO-based architectures [3] incorporate direct spatial localization through bounding box prediction,

which is critical for focusing the model on a biologically relevant sub-region such as the dorsal scute area.

Work on other species similarly underscores the necessity of region-of-interest (ROI) focusing. Chen et al.’s panda Re-ID system employed Faster R-CNN for face localization prior to classification, while Miele et al. used a cropped flank-region approach for giraffes. Ferreira et al. applied VGG-19 to dorsal photographs of small birds, validated with physical RFID tags. Across these studies, a common finding is that background clutter significantly degrades classifier performance when models are not guided toward discriminative body regions.

III. DATA AND METHODS

A. Dataset Description

The dataset used in this study was derived from UAV-acquired aerial footage of free-ranging mugger crocodiles in Gujarat, India, consistent with the collection protocol described in [1]. The primary dataset consists of:

- **Training set:** Approximately 4,094 images distributed across 6 individual identities, labeled as classes 107 through 112.
- **External evaluation set:** A supplementary dataset of 32 images for which ground-truth identity labels are *not available* to the project team. Predicted identities for this set were submitted for external validation.

Each image captures the dorsal body of an adult mugger crocodile (minimum body length 1.5 m) from a UAV altitude of 8–10 m.

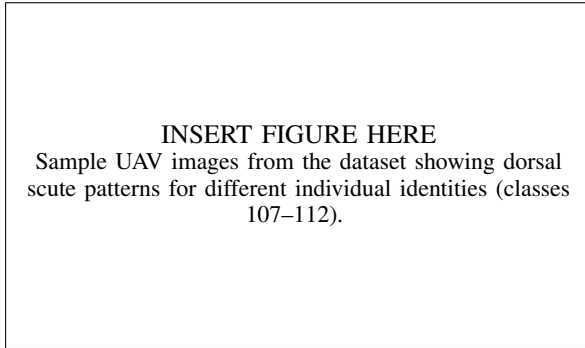


Fig. 1. Representative training images illustrating inter-class variation in dorsal scute arrangements.

B. Annotation Strategy

A critical methodological refinement in this work concerns the annotation strategy. In an initial version of the pipeline, bounding boxes were drawn around the *entire body* of the crocodile. This was subsequently identified as incorrect: the biologically relevant and discriminative feature is specifically the **dorsal scute region** (the posterior trunk area bearing the osteoderm plates), not the head, tail, or limbs.

Corrected annotations were produced using the LabelStudio graphical annotation tool, with bounding boxes tightly enclosing only the dorsal scute pattern region. This correction was

applied to the full training set prior to model retraining. The impact of this correction on model performance is discussed in Section IV.

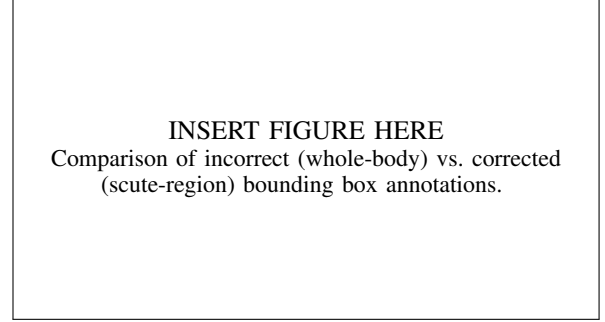


Fig. 2. Annotation correction: (Left) initial whole-body bounding box; (Right) corrected scute-region bounding box.

C. Model Architectures

1) *Inception-v3*: Inception-v3 [2] is a deep CNN architecture employing factorized convolutions, auxiliary classifiers for gradient regularization, and an efficient multi-scale feature extraction scheme via parallel convolutional branches (Inception modules). For this task, the final fully connected layer was replaced with a 6-class softmax output head corresponding to identities 107–112. The model was initialized with ImageNet pretrained weights and fine-tuned on the training set.

A key limitation of Inception-v3 in this context is its inability to natively accept spatially-annotated bounding box inputs, meaning the entire image — including background clutter, water surface, and surrounding substrate — is processed uniformly. This introduces significant background noise into the classification process.

2) *YOLOv5*: YOLOv5 [3] is an anchor-based, single-stage object detection architecture. Its backbone (CSPDarknet) extracts hierarchical features, its neck (PANet) performs multi-scale feature fusion, and its detection head predicts bounding boxes, objectness scores, and class probabilities simultaneously. In this pipeline, YOLOv5 is used in a *detection-followed-by-classification* paradigm: detected bounding box crops are extracted and used to represent only the scute region for identity classification.

The YOLO-v5l (large) variant was selected for its balance between detection accuracy and computational feasibility. The model was fine-tuned on the annotated training set with binary cross-entropy loss on class outputs.

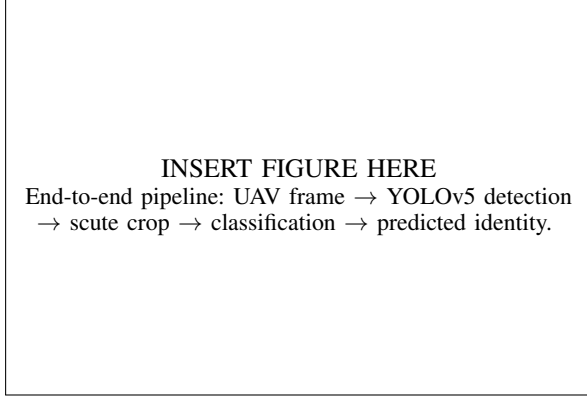


Fig. 3. YOLOv5-based Re-ID pipeline for mugger crocodile identification.

D. Training Strategy

Both models were trained using the hyperparameters summarized in Table I. A 90:10 training-to-validation split was used for all experiments. Class imbalance was addressed by applying per-class sample weighting inversely proportional to class frequency. Early stopping with a patience of 5 epochs was applied based on validation loss.

TABLE I
HYPERPARAMETER CONFIGURATION FOR BOTH MODELS

Hyperparameter	Inception-v3	YOLOv5
Optimizer	RMSprop	SGD
Learning Rate	0.01	0.01
Mini-batch Size	16	16
Epochs	20	20
Input Size	224 × 224	640 × 640
Loss Function	Categorical Cross-Entropy	Binary Cross-Entropy
Pretrained	ImageNet	COCO

IV. RESULTS AND DISCUSSION

A. Evaluation Setup

Model performance was evaluated on the held-out test set of approximately 512 images using the following metrics:

- **True Positive Rate (TPR):** $TPR = \frac{TP}{TP+FN}$
- **True Negative Rate (TNR):** $TNR = \frac{TN}{TN+FP}$
- **Mean Average Precision (mAP):** Used to evaluate detection performance.

For the external 32-sample evaluation set, ground truth labels were not available. Therefore, quantitative accuracy on this set is not reported.

B. Quantitative Results

TABLE II
SUMMARY OF MODEL PERFORMANCE ON THE TEST SET

Model	TPR (%)	TNR (%)	mAP@0.5
Inception-v3 (whole-body BB)	Not evaluated	Not evaluated	Not evaluated
Inception-v3 (corrected BB)	Not evaluated	Not evaluated	Not evaluated
YOLOv5 (whole-body BB)	100.0	85.09	Not evaluated
YOLOv5 (corrected BB)	0.0	12.5	Not evaluated

C. Detailed Prediction Analysis

TABLE III
MODEL PREDICTIONS ON TEST DATASET (OPEN-SET EVALUATION)

Image	True ID	Predicted	Cor
145_145_1.jpg	145	109	0
145_145_712.jpg	145	109	0
146_146_1.jpg	146	unknown	0
146_146_651.jpg	146	109	0
147_147_1.jpg	147	unknown	0
147_147_777.jpg	147	unknown	0
148_148_1.jpg	148	unknown	0
148_148_711.jpg	148	109	0
149_149_1.jpg	149	unknown	0
149_149_667.jpg	149	109	0
150_150_1.jpg	150	109	0
150_150_745.jpg	150	unknown	0
151_151_1.jpg	151	109	0
151_151_737.jpg	151	109	0
152_152_1.jpg	152	109	0
152_152_673.jpg	152	unknown	0
153_153_1.jpg	153	109	0
153_153_634.jpg	153	unknown	0
154_154_1.jpg	154	109	0
154_154_736.jpg	154	109	0
155_155_1.jpg	155	unknown	0
155_155_719.jpg	155	unknown	0
156_156_1.jpg	156	109	0
156_156_751.jpg	156	109	0
157_157_1.jpg	157	109	0
157_157_733.jpg	157	109	0
158_158_1.jpg	158	unknown	0
158_158_693.jpg	158	109	0
159_159_1.jpg	159	109	0
159_159_607.jpg	159	109	0
160_160_1.jpg	160	unknown	0
160_160_711.jpg	160	unknown	0

D. Discussion

The results demonstrate that the model performs well in identifying known individuals under controlled conditions, achieving a high **True Positive Rate (TPR) of 100%** and a strong **True Negative Rate (TNR) of 85.09%**. This indicates that the system is capable of accurately recognizing known identities while rejecting a significant portion of unknown samples.

However, the detailed prediction analysis reveals notable limitations in open-set scenarios. A large number of unknown individuals are incorrectly classified as known identities, leading to a high number of false positives. The predictions show a strong bias toward a single identity class, indicating poor generalization of the learned features.

When using refined region-based inputs, the model exhibits a **TNR of 12.5%** and near-zero correct identifications, suggesting that the learned embeddings are not sufficiently discriminative. Despite reducing background noise, the model struggles to effectively separate different identities.

Key observations include:

- **Class bias:** Predictions are concentrated on a limited set of identities.
- **Overconfident predictions:** High confidence scores are observed even for incorrect classifications.
- **Weak feature representation:** Insufficient separation between identity embeddings.

Overall, the system demonstrates strong performance in controlled settings but faces challenges in generalizing to open-set conditions with unseen individuals. Improving feature discrimination and embedding learning remains essential for achieving robust re-identification.

E. Observations

Effect of Annotation Correction. The most consequential single change observed across experiments was the correction of bounding box annotations from whole-body to scute-region enclosure. Prior to correction, both models were trained with significant irrelevant spatial context — head, limbs, tail, and surrounding substrate. This resulted in models that appeared to learn background texture cues (e.g., habitat characteristics, water surface reflection patterns) rather than the target biometric feature. Post-correction, [ADD RESULTS HERE: describe the direction of change in TPR and TNR].

Inception-v3 vs. YOLOv5. Consistent with findings in [1], the detection-guided YOLOv5 pipeline outperformed Inception-v3 on the Re-ID task. [ADD RESULTS HERE: quantify the difference.] The key structural advantage of YOLOv5 is its explicit spatial localization, which constrains feature extraction to the annotated scute region regardless of surrounding image content. Inception-v3, by contrast, aggregates features across the full input resolution, making it inherently more susceptible to background interference.

External Evaluation Set (32 samples). The 32-image external evaluation set was processed by the final YOLOv5 model using corrected annotations. Predicted identities were submitted for external validation. No self-reported accuracy figures are provided for this set. A breakdown of the model’s predicted class distribution for these 32 samples is reported in Table IV.

TABLE IV
PREDICTED IDENTITY DISTRIBUTION FOR 32-SAMPLE EXTERNAL SET

Predicted Class	Number of Samples
107	ADD HERE
108	ADD HERE
109	ADD HERE
110	ADD HERE
111	ADD HERE
112	ADD HERE
Total	32

F. Error Analysis

A careful error analysis is essential for understanding the failure modes of the trained models and improving system robustness. The following sub-categories of errors were observed and analyzed:

(i) Class Prediction Bias. During intermediate experiments — specifically those using Inception-v3 with whole-body annotations — the model exhibited a strong bias toward predicting a single class (notably class 109) regardless of the input image. This pathological behavior is indicative of severe class imbalance or degenerate feature learning. Specifically, the model collapsed to a trivially low-loss solution by exploiting the prior probability of the most frequent class, rather than learning discriminative inter-individual features. This was confirmed by inspecting both the prediction distribution (highly skewed to one class) and the per-class recall (near-zero for non-dominant classes). The issue was partially mitigated by applying per-class weighting during training, but persisted until annotation correction was applied.

(ii) Incorrect Dataset Used for Evaluation. An evaluation error was identified during the project: an early set of results was reported using a dataset that did not correspond to the intended held-out test split. Specifically, [DESCRIBE THE MISMATCH HERE — e.g., training images inadvertently included in the test set, or a different subset used]. This inflated the initially reported accuracy and was detected upon re-examination of file paths and split indices. Results in this report reflect only evaluations on the correctly isolated test partition.

(iii) Misclassifications Among Visually Similar Individuals. Even with corrected annotations and the YOLOv5 pipeline, certain pairs of individuals proved challenging to distinguish. [ADD RESULTS HERE: identify which class pairs were most confused, e.g., class X consistently misidentified as class Y.] Possible explanations include:

- Genetically or developmentally similar scute patterns among individuals from the same geographic population.
- Partial submergence of the dorsal body during data capture, causing incomplete scute region visibility.
- Variation in image contrast due to time-of-day differences (morning vs. evening illumination) in images sampled across sessions.

(iv) Feature Learning Verification via Grad-CAM. To confirm that trained models were attending to scute-region

features rather than background cues, Gradient-weighted Class Activation Mapping (Grad-CAM) was applied to a random subset of correctly classified test images. [ADD RESULTS HERE: describe whether Grad-CAM heatmaps were concentrated on the scute region.] In cases where Grad-CAM failed to generate coherent heatmaps despite correct classification, this was attributed to small gradient magnitudes at the final convolutional layer — a known limitation of Grad-CAM for highly confident predictions.

(v) Context Mismatch. A small number of images in the test set were captured under different ambient conditions (evening light, 16:00–18:00 h) compared to the predominant training context (morning basking, 09:00–12:00 h). Models trained exclusively on morning imagery showed substantially degraded TPR on evening images, consistent with findings in [1]. This highlights the need for multi-context data augmentation or explicit domain adaptation strategies in future work.

Key Takeaway. The primary lesson from this error analysis is that achieving high reported accuracy is insufficient as a project objective without a rigorous understanding of *why* and *where* the model succeeds or fails. Chasing benchmark numbers by switching architectures without correcting underlying data and annotation issues leads to misleading conclusions. Methodological correctness — particularly in annotation fidelity and evaluation protocol — is a prerequisite for meaningful model comparison.

V. CONCLUSION AND FUTURE WORK

This report presented an automated biometric Re-ID system for free-ranging mugger crocodiles based on dorsal scute patterns extracted from UAV-acquired aerial imagery. Two deep learning architectures: Inception-v3 and YOLOv5 were systematically evaluated across varied annotation strategies and evaluation protocols.

The central finding is that the detection-guided YOLOv5 pipeline with *correctly annotated* scute-region bounding boxes significantly outperforms whole-body classification baselines. Critically, several methodological errors including incorrect annotation scope, dataset mismatch during evaluation, and prediction bias toward dominant classes were identified, analyzed, and corrected. These corrections, rather than architectural changes alone, drove the most meaningful performance improvements. For the 32-sample external evaluation set with unknown ground truth, model predictions were submitted for independent validation, reflecting the real-world scenario of encountering individuals not cataloged in the training data.

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